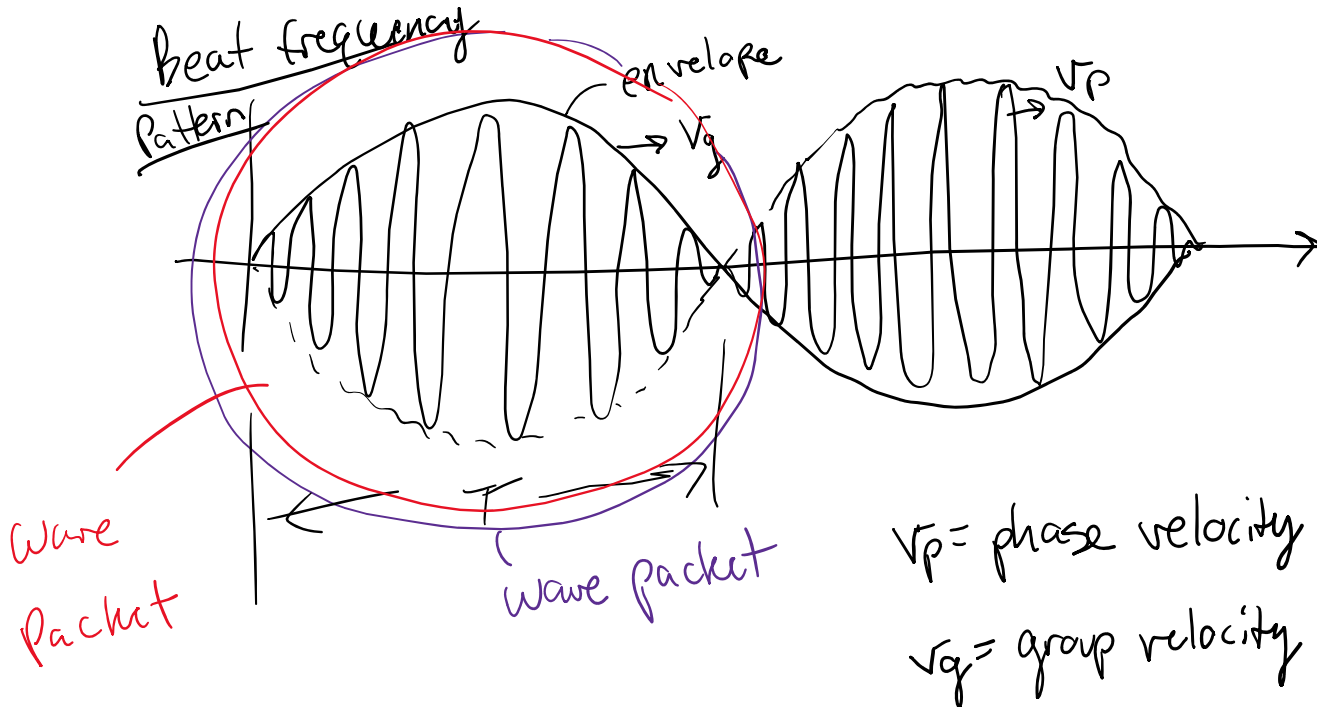


$$E_{XT}(z, t) = 2E_0 \cos[(\Delta\omega)t - (\Delta k)z] \cos(\omega t - kz)$$



Define E_{max} →

$$[(\Delta\omega)t - (\Delta k)z] = 2\pi m; \quad m \text{ is an integer}$$

Wave packet travels at the group velocity

$$\frac{\Delta z}{\Delta t} = \frac{\Delta\omega}{\Delta k} = (\text{Group velocity}) v_g$$

Wave Packet Period

$$T = \frac{2\pi}{\Delta\omega}$$

in freespace (non-dispersive medium)

$$\frac{\Delta\omega}{\Delta k} = \text{constant} = c \left(3 \times 10^8 \frac{\text{m}}{\text{s}} \right)$$

* only in freespace $v_g = v_p$

In a medium

$n = \sqrt{\epsilon_r}$; $n = \text{refractive index}$

$$k = \frac{2\pi n}{\lambda_0}$$

freespace

$$v_p = \frac{\omega}{k} = \frac{\omega \lambda_0}{2\pi n}$$

$$v_p = \frac{f \lambda_0}{n}$$

$$v_g = \frac{\Delta\omega}{\Delta k} = \frac{d\omega}{dk} = \left(f \frac{d\lambda_0}{dn} \leftrightarrow f \frac{d\lambda_0}{dn} \right)$$

same

Define a "Group" index of Refraction

- how the 'group' pattern interacts with the electrical spring constant (permittivity $\rightarrow$ Relative ϵ_r)

$$N_g = n - \lambda_0 \left(\frac{dn}{d\lambda_0} \right) \rightarrow \text{dispersion term}$$

a situation could arise where the dispersion equals and is canceled by a non-linear change in the medium created by the magnitude of the wave as it travels thru.

New wave (Soliton wave)

$$\vec{E} = - \left(\frac{1}{2} v \right) \text{sech}^2 \left[\frac{\sqrt{v}}{2} (x - vt \pm \theta) \right]$$

$\uparrow$
amplitude.